Mathematics - Undergraduate Level Assessment 11 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Statement 1 | If H is a subgroup of a group G and a belongs to G, then aH = Ha. Statement 2 | If H is normal of G and a belongs to G, then ah = ha for all h in H.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
2. Compute the product in the given ring. (12)(16) in Z_24
   (a) 0
   (b) 1
   (c) 4
   (d) 6
3. For T: Z x Z -> Z where T(1, 0) = 3 and T(0, 1) = -5, find T(-3,2).
   (a) -19
   (b) -10
   (c) 19
   (d) 10
4. Find the characteristic of the ring Z_3 x Z_3.
   (a) 0
   (b) 3
   (c) 12
   (d) 30
5. Statement 1 | Any set of two vectors in R^2 is linearly independent. Statement 2 | If V = span(v 1, ... , vk) and {v 1, ... , vk} are linearly independent, then dim(V) = k.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True

True / False (10 marks)

Answer True or False

6. True or False: In the finite field Z_7, the element 2 is a generator of the multiplicative group of nonzero elements.

7. True or False: If$_3$ x + 7 y is divisible by 11, then 9 x + 4 y must also be divisible by 11 for positive integers x and y.

8. True or False: In the group G = {2, 4, 6, 8} under multiplication modulo 10, the identity element is 6.

9. True or False: It is possible to have a nontrivial homomorphism from a finite group into an infinite group.

10. True or False: If a and b are elements of finite order in an Abelian group, then |ab| is the least common multiple of |a| and |b| under certain conditions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. A circle centered at A with a radius of 1 and a circle centered at B with a radius of 4 are externally tangent. A third circle is tangent to the first two and to one of their common external tangents as shown. What is the radius of the third circle? [asy] draw((-3,0)–(7.5,0)); draw(Circle((-1,1),1),linewidth(0.7)); draw(Circle((3,4),4),linewidth(0.7)); draw(Circle((0.33 dot((-1,1)); dot((3,4)); draw((-1,1)–(-2,1)); draw((3,4)–(7,4)); label("A",(-1,1),E); label("B",(3,4),W); label("1",(-1.5,1),N); label("4",(5,4),N); [/asy]

12. Discuss the conditions under which a triangular region, with one vertex at the center of a circle of radius 1 and the other two vertices on the circle, achieves its maximum area. Calculate this area and justify your reasoning.

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